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10 CFR 50.73

September 29, 2020
Serial: RA-20-0269

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555

Subject: Brunswick Steam Electric Plant, Unit No. 1
Renewed Facility Operating License No. DPR-71
Docket No. 50-325
Licensee Event Report 1-2020-004

In accordance with the Code of Federal Regulations, Title 10, Part 50.73, Duke Energy Progress, LLC, is submitting the enclosed Licensee Event Report (LER). This report fulfills the requirement for a written report within sixty (60) days of a reportable occurrence.

This document contains no regulatory commitments.

Please refer any questions regarding this submittal to Ms. Sabrina Salazar, Manager – Nuclear Support Services, at (910) 832-3207.

Sincerely,

John A. Krakuszeski

SBY/sby

Enclosure: Licensee Event Report

cc (with enclosure):

Ms. Laura Dudes, NRC Regional Administrator, Region II
Mr. Andrew Hon, NRC Project Manager
Mr. Gale Smith, NRC Senior Resident Inspector
Chair - North Carolina Utilities Commission

NRC FORM 366 (08-2020)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 08/31/2023

LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
Brunswick Steam Electric Plant (BSEP), Unit 1	05000325	1 OF 3

4. Title
Setpoint Drift in Main Steam Line Safety Relief Valves Results in Three Valves Inoperable

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
08	12	2020	2020	- 004 -	00	09	29	2020		05000
									Facility Name	Docket Number 05000

9. Operating Mode	10. Power Level
4	000

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	[ ] 20.2203(a)(2)(vi)	[ ] 50.36(c)(2)	[ ] 50.73(a)(2)(iv)(A)	[ ] 50.73(a)(2)(x)
[ ] 20.2201(b)	[ ] 20.2203(a)(3)(i)	[ ] 50.46(a)(3)(ii)	[ ] 50.73(a)(2)(v)(A)	**10 CFR Part 73**
[ ] 20.2201(d)	[ ] 20.2203(a)(3)(ii)	[ ] 50.69(g)	[ ] 50.73(a)(2)(v)(B)	[ ] 73.71(a)(4)
[ ] 20.2203(a)(1)	[ ] 20.2203(a)(4)	[ ] 50.73(a)(2)(i)(A)	[ ] 50.73(a)(2)(v)(C)	[ ] 73.71(a)(5)
[ ] 20.2203(a)(2)(i)	**10 CFR Part 21**	[X] 50.73(a)(2)(i)(B)	[ ] 50.73(a)(2)(v)(D)	[ ] 73.77(a)(1)(i)
[ ] 20.2203(a)(2)(ii)	[ ] 21.2(c)	[ ] 50.73(a)(2)(i)(C)	[ ] 50.73(a)(2)(vii)	[ ] 73.77(a)(2)(i)
[ ] 20.2203(a)(2)(iii)	**10 CFR Part 50**	[ ] 50.73(a)(2)(ii)(A)	[ ] 50.73(a)(2)(viii)(A)	[ ] 73.77(a)(2)(ii)
[ ] 20.2203(a)(2)(iv)	[ ] 50.36(c)(1)(i)(A)	[ ] 50.73(a)(2)(ii)(B)	[ ] 50.73(a)(2)(viii)(B)	
[ ] 20.2203(a)(2)(v)	[ ] 50.36(c)(1)(ii)(A)	[ ] 50.73(a)(2)(iii)	[ ] 50.73(a)(2)(ix)(A)	

[ ] **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Sabrina Salazar, Manager – Nuclear Support Services	(910) 832-3207

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS		Cause	System	Component	Manufacturer	Reportable to IRIS
D	SB	RV	T020	Y						

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
[X] No	[ ] Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On August 12, 2020, BSEP received test results for the eleven main steam line safety relief valves (SRVs) removed from Unit 1 during the spring 2020 refuel outage (RFO). Three of the eleven valves were found to have as found pilot valve lift setpoints outside the +/-3 percent tolerance required by Technical Specification (TS) 3.4.3, which requires ten of the eleven installed SRVs to be operable. Since less than ten SRVs were operable, this event is being reported per 10 CFR 50.73(a)(2)(i)(B) as operation prohibited by the plant's TS. At the time the test results were received, Unit 1 was shutdown for an unrelated maintenance outage.

The direct cause of the out of tolerance as-found lift setpoints was corrosion bonding of the SRV pilot discs to the pilot seats. The root cause of the corrosion bonding was inadequate procedures in place when these valves were refurbished and installed in the 2018 RFO. Subsequent to the 2018 installation of these SRVs, as a result of the Root Cause Evaluation associated with LER 1-2018-003, dated August 9, 2018 (i.e., ADAMS Accession Number ML18221A552), corrective actions were completed revising the associated procedures to prevent this condition from recurring.

All eleven of the SRV pilot valves were replaced with certified spares before Unit 1 startup from the spring 2020 RFO. These spares were rebuilt utilizing the revised procedures.

NARRATIVE

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].

Background

Initial Conditions

At the time of the event, Unit 1 was in Mode 4 (i.e., Cold Shutdown), for an unrelated maintenance outage.

Reportability Criteria

Unit 1 Technical Specification (TS) 3.4.3 requires at least ten of eleven main steam [SB] line safety relief valves (SRVs) to be operable. Per Surveillance Requirement (SR) 3.4.3.1, each valve is required to open within +/- 3 percent of its opening setpoint. As-found testing of the valves indicated that three of the valves had lift setpoints outside this tolerance. Consequently, the plant operated in a condition which is prohibited by the TS, that is, with fewer than the required number of SRVs having lift setpoints within the +/-3 percent tolerance. Therefore, the condition is being reported per 10 CFR 50.73(a)(2)(i)(B) as operation in a condition prohibited by the plant TSs.

Event Description

During the spring 2020 Unit 1 refueling outage, all eleven pilot valve assemblies in the SRVs were replaced with certified spares. The removed SRV pilot valves were sent to National Technical Systems (NTS) to determine the as-found set pressure. On August 12, 2020, test results were reported to BSEP. The test results showed that three of the eleven valves actuated at pressures outside of the +/-3 percent tolerance allowed by TS 3.4.3. Specifically, all three of the valves that lifted outside of the tolerance, lifted at greater than +3%.

Event Cause

The change in SRV lift setpoints resulted from corrosion bonding of the SRV pilot discs-to-the pilot seats. Corrosion bonding between the pilot disc and seat is an inherent problem with the two-stage SRV design used at BSEP. BSEP mitigates corrosion bonding of SRVs by coating each pilot disc surface with platinum. The integrity of the platinum coating is critical to mitigating corrosion bonding. The platinum coating of these SRVs was found to be degraded in large areas of the pilot disc, including the seating surface; which was determined to be caused by circumferential machining marks in the substrate surface that led to increased circumferential cracking of the platinum coating.

The root cause was attributed to the procedures in place when these valves were refurbished and installed in the 2018 refueling outage lacking sufficient detail to ensure consistent surface preparation and proper quality checks of surface condition prior to platinum coating.

Safety Assessment

The purpose of the SRVs is to provide overpressure protection for the reactor coolant system. The as-found condition of the Unit 1 SRVs was compared to cases analyzed in the Brunswick Unit 1 Cycle 22 Reload Safety Analysis Report (RSAR). Each as-found SRV was bounded by the opening pressures assumed in the analysis; therefore, the overall RSAR analysis bounds the overall as-found condition of the SRVs.

The results from the RSAR analysis demonstrate that the pressure limits are not exceeded for the overpressurization events. Therefore, it is concluded that the Unit 1 Cycle 22 SRVs always remained capable of performing their safety function of preventing overpressurization of the reactor coolant system.

Based on the foregoing analysis, it is concluded that this event had no adverse impact on nuclear safety.

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Brunswick Steam Electric Plant (BSEP), Unit 1	05000- 325	2020	- 004	- 00

NARRATIVE

Corrective Actions

Subsequent to the 2018 installation of these SRVs, as a result of the Root Cause Evaluation (RCE) associated with LER 1-2018-003, dated August 9, 2018 (i.e., ADAMS Accession Number ML18221A552), corrective actions were completed revising the associated procedures to add the necessary guidance to ensure consistent surface preparation and proper quality checks of surface condition prior to platinum coating, to prevent this condition from recurring. All eleven of the SRV pilot valves were replaced with certified spares before Unit 1 startup from the spring 2020 refueling outage. The certified spares were rebuilt utilizing the revised procedures.

In addition to the aforementioned completed corrective actions, BSEP currently plans to pursue a license amendment request to adopt Technical Specification Task Force (TSTF) 576, following its approval by the NRC, to align TS 3.4.3 with the safety function of the SRVs vs. the currently prescribed +/-3 percent setpoint tolerance.

Previous Similar Events

A review of LERs for the past three years identified the following previous similar event associated with SRVs with as-found lift setpoints outside of TS allowable limits.

- LER 1-2018-003, dated August 9, 2018, reported Unit 1 operation prohibited by TSs due to two of the eleven valves found with lift setpoints of their pilot valves outside the +/-3 percent tolerance required by TS 3.4.3.

The direct cause of this event was corrosion bonding. The root cause was attributed to the procedures lacking sufficient detail to ensure consistent surface preparation and proper quality checks of surface condition prior to platinum coating.

Commitments

No regulatory commitments are contained in this report.